

Assessment of Towing Devices for Force Measurement in Elite Level Swimming

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INTRODUCTION

World record swimming performances to date have been a product of biomechanical refinement, built on the understanding of propulsive phases and drag minimisation strategies [1]. At the elite level, the initial push off-the-wall velocity peaks at 2.2-1.9 m/s, then tapers off within 0.5 seconds, before propulsive underwater kicks commence [2, 3]. In a long course 50 metre freestyle event, the underwater phases can take up to 30% of the total race distance. [4] There is a challenge of identifying tools to measure passive drag within the complexities of the swimming environment. Motorized tethered devices (MTDs) have emerged with an ability to tow swimmers under passive drag conditions; MTDs such as the 1080 Motion Sprint (Version 1, 1080 Motion AB, Stockholm, Sweden), and T-APEX (Hunting Technology Limited, Hong Kong, China) are used widely across training environments. These devices have not yet been tested for reliability. This study aims to: (1) assess the reliability of the 1080 and T-APEX MTD's in measuring hydrodynamic drag of a mannequin and a national level swimming cohort; (2) assess hydrodynamic drag experienced by swimmers under two drag conditions (low-, and high-drag) and determine the level of agreement between the MTDs.

METHODS

Twelve Australian senior national level able swimmers [5], (7 Male, 5 Female, aged 18 and above) were recruited from high-performance clubs around Melbourne (Victoria, Australia). A custom-built steel frame, clamped down to pool deck, allowed for the MTD's tether to tow the subject at a desired depth of 0.8 metres, during each trial. The study included two testing stages. The first stage involved horizontal towing of a mannequin (1:2 scale), in an athlete-independent condition; which aimed to mitigate surface drag interference. The mannequin was three dimensionally printed with polymer to replicate an unpublished mannequin model developed by Swimming Australia to depict an optimal streamline position. The MTD towed the mannequin across a pre-determined 12-metre segment, for 10 trials at 2.3 m/s and 5.0 m/s. The second stage involved the hydrodynamic drag assessment of human participants. Each participant was required to be towed under two different conditions at three pre-determined velocities (1.5 m/s, 1.9 m/s, and 2.3 m/s) within each condition. The low drag (LD) condition required participants to be towed wearing their own World Aquatics approved race suit. The high drag (HD) condition required participants to be towed wearing their own World Aquatics approved race suit, with a parachute attached to their waist.

RESULTS AND DISCUSSION

Force measurement during mannequin testing presented poor agreement between devices at 2.3m/s (ICC = 0.080, F = 1.37, p= 0.387) and moderate agreement, with no statistical significance at 5.0m/s (ICC = 0.410, F= 1.93, p= 0.340). Within-device reliability of force measurement for the 1080 and T-Apex when towing swimmers, had mixed results. The Intra Class Correlation showed statistically significant reliability in both LD (1080: ICC = 0.833, F = 11.0, p = 0.002; T-APEX: ICC = 0.812, F = 9.59, p = .002) and HD conditions at 1.5m/s (1080: ICC = 0.747, F = 6.49, p = 0.010; T-APEX: ICC = 0.797, F = 14.1, p = 0.016). Reliability was higher again at 1.9 m/s for swimmers; both devices suggested excellent reliability across the LD condition (1080: ICC = 0.944, F = 41.4, p= 0.000; T-APEX: ICC = 0.874, F = 102, p = .000). Reliability was very low (i.e. not significant) in either device when towed at 2.3m/s. This could suggest swimmers were less consistent in holding streamline at higher speeds, thus creating discrepancies across trials.

There was poor agreement between the 1080 and T-APEX devices across all speeds and drag conditions, with no ICC values greater than 0.0. Separately, force data from the HD condition at both 1.5m/s and 1.9m/s achieved a negative ICC value (ICC -0.469 and ICC -0.221, respectively) suggesting that within-swimmer variance between the 1080 and T-APEX, was greater than the variance across swimmers.

CONCLUSIONS

The 1080 and T-APEX can reliably perform assisted towing on swimmers at 1.5m/s and 1.9m/s, under LD conditions. However, force measurement data must be analysed within one device, as they produce incomparable force outputs when towing swimmers at the same speeds. ICC values for between-device reliability showed no agreement.

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